

Formula Sheet

Useful facts about Differential Equations

The DE $y' = ay$ has solution of the form $y = Ae^{ax}$

The DE $y' = yf(x)$ has solution of the form $y = Ae^{\int f(x) dx}$

The Simple Harmonic Oscillator DE $y'' + \omega^2 y = 0$ has solutions of the form

$$y = A \cos(\omega x) + B \sin(\omega x)$$

The DE $y'' - k^2 y = 0$ has solutions of the form

$$y = A \cosh(kx) + B \sinh(kx)$$

Vector Calculus in Cartesian Coordinates (x, y, z)

Curves C in space represented by the position vector $\vec{r}(t)$ have $\vec{r}'(t)$ as a tangent vector.

Common parameterizations for curves:

$$\text{Straight Line : } x(t) = at + x_0, \quad y(t) = bt + y_0$$

$$\text{Circle : } x(t) = R \cos(t), \quad y(t) = R \sin(t)$$

$$\text{Ellipse : } x(t) = a \cos(t), \quad y(t) = b \sin(t)$$

$$\text{Circular Helix : } x(t) = R \cos(t), \quad y(t) = R \sin(t), \quad z(t) = ct$$

$$\text{Gradient of Scalar field } f: \nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$$

Directional Derivative of f in the direction of a vector $\vec{a}$:

$$D_{\vec{a}} f = \frac{1}{|\vec{a}|} \vec{a} \cdot \nabla f$$

Divergence and curl of vector function $\vec{v} = [v_1, v_2, v_3]$:

$$\nabla \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$\nabla \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Basic properties of grad, div, and curl:

$$\nabla(fg) = f\nabla g + g\nabla f$$

$$\nabla \cdot (f\vec{v}) = f(\nabla \cdot \vec{v}) + \vec{v} \cdot \nabla f$$

$$\nabla^2 f = \nabla \cdot (\nabla f)$$

$$\nabla \times (f\vec{v}) = \nabla f \times \vec{v} + f(\nabla \times \vec{v})$$

$$\nabla \times (\nabla f) = 0$$

$$\nabla \cdot (\nabla \times \vec{v}) = 0$$

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General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D|$$

$$\text{Volume element: } dV = \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} dq_1 dq_2 dq_3.$$

Orthogonal Coordinates

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const} \ (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right].$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

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Partial Differential Equations

The one-dimensional wave equation $u_{tt} = c^2 u_{xx}$ with initial conditions $u(x, 0) = f(x)$, $u_t(x, 0) = 0$ for $-\infty < x < \infty$, and no specific boundary conditions, has solution based on D'Alembert's work:

$$u(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)].$$

The one-dimensional wave equation $u_{tt} = c^2 u_{xx}$ with standing wave boundary conditions $u(0, t) = 0 = u(L, t)$ for $t \geq 0$ and $L > 0$, and initial conditions $u(x, 0) = f(x)$, $u_t(x, 0) = g(x)$ for $0 \leq x \leq L$, has solution based on Fourier Series given by

$$u(x, t) = \sum_{n=1}^{\infty} \left(A_n \cos\left(\frac{n\pi c}{L}t\right) + B_n \sin\left(\frac{n\pi c}{L}t\right) \right) \sin\left(\frac{n\pi}{L}x\right),$$

with $A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$, and $B_n = \frac{2}{nc\pi} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx$.

The one-dimensional heat equation $u_t = c^2 u_{xx}$ with boundary conditions $u(0, t) = u(L, t) = 0$ for $t \geq 0$ and initial condition $u(x, 0) = f(x)$ has solution based on the Fourier Series given by

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L}x\right) e^{-\left(\frac{cn\pi}{L}\right)^2 t}$$

with $A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$.

Two dimensional wave equation $u_{tt} = c^2 \nabla^2 u$ on a rectangular membrane $0 \leq x \leq a$, $0 \leq y \leq b$ with $u = 0$ on the boundary and initial conditions $u(x, y, 0) = f(x, y)$, $u_t(x, y, 0) = g(x, y)$ has solution based on the double Fourier Series:

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (A_{mn} \cos(\lambda_{mn}t) + B_{mn} \sin(\lambda_{mn}t)) \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right),$$

where $\lambda_{mn} = c\pi \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$, and

$$A_{mn} = \frac{4}{ab} \int_0^b \int_0^a f(x, y) \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) dx dy$$

$$B_{mn} = \frac{4}{ab\lambda_{mn}} \int_0^b \int_0^a g(x, y) \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) dx dy$$

On a circular membrane of radius R , assuming $u = u(r, t)$ is independent of the angle θ and $u = 0$ on the boundary, the solutions to the wave equation $u_{tt} = c^2 \nabla^2 u$ with initial conditions $u(r, 0) = f(r)$ and $u_t(r, 0) = 0$, can be written in terms of the the Bessel-Fourier series as:

$$u(r, t) = \sum_{m=1}^{\infty} A_m \cos(\lambda_m t) J_0\left(\frac{\alpha_m}{R}r\right).$$

Here α_m are the roots of the Bessel function $J_0(s)$. If $f(r)$ is expressed as

$$f(r) = \sum_{k=i}^j c_k J_0\left(\frac{\alpha_k}{R}r\right),$$

for some positive integers $i < j$, then for the solution, $A_i = c_i$, $A_{i+1} = c_{i+1}, \dots, A_j = c_j$, and $A_m = 0$ for any m that is not between i and j .

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Fourier Transforms

Definition: $\hat{u}(\omega, t) = \mathcal{F}(u) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u(x, t) e^{-i\omega x} dx$

Derivatives: $\mathcal{F}(u_t) = \frac{\partial \hat{u}}{\partial t}$, and $\mathcal{F}(u_x) = -i\omega \hat{u}$, so $\mathcal{F}(u_{tt}) = \frac{\partial^2 \hat{u}}{\partial t^2}$ and $\mathcal{F}(u_{xx}) = -\omega^2 \hat{u}$.

If $u(x, 0) = f(x)$ then $\hat{u}(\omega, 0) = \hat{f}(\omega)$, where $\hat{f} = \mathcal{F}(f) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$.

If $\hat{f} = \mathcal{F}(f)$ and $\hat{g} = \mathcal{F}(g)$, then

$$\int_{-\infty}^{\infty} \hat{f} \cdot \hat{g} \cdot e^{i\omega x} d\omega = (f * g)(x) = \int_{-\infty}^{\infty} f(p)g(x-p) dp.$$

Laplace Transforms

Definition: $F(s) = \mathcal{L}[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$, and $f(t) = \mathcal{L}^{-1}[F(s)]$

Linearity: $\mathcal{L}[af(t) + bg(t)] = aF(s) + bG(s)$

Derivatives:

$$\begin{aligned}\mathcal{L}(f') &= sF(s) - f(0) \\ \mathcal{L}(f'') &= s^2 F(s) - sf(0) - f'(0)\end{aligned}$$

Shifting (here $u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$ is the Heaviside function.

$$\begin{aligned}\mathcal{L}[e^{at} f(t)] &= F(s-a) \\ \mathcal{L}^{-1}[e^{-as} F(s)] &= f(t-a)u(t-a) = \begin{cases} f(t-a) & \text{if } t \geq a \\ 0 & \text{if } t < a. \end{cases}\end{aligned}$$

Specific formulae:

$$\begin{aligned}\mathcal{L}[1] &= \frac{1}{s} \\ \mathcal{L}[t] &= \frac{1}{s^2} \\ \mathcal{L}\left[\frac{t^{n-1}}{(n-1)!}\right] &= \frac{1}{s^n} \\ \mathcal{L}[e^{at}] &= \frac{1}{s-a} \\ \mathcal{L}[u(t-a)] &= \frac{e^{-as}}{s}\end{aligned}$$